AgriMind

Agriculture Knowledge Base

A consolidated, source-linked reference for sustainable farming, crop care, soil health, pest and disease management, and weather-aware decisions.

Prepared for: AgriMind — Multimodal AI Assistant for Sustainable Agriculture

Regional focus: Tamil Nadu, India | **Version:** 1.0 | **Prepared:** 14 October 2026

This is a curated synthesis with links to official source material—not a verbatim merger of every source publication. It is intended as a starter PDF for a Retrieval-Augmented Generation (RAG) knowledge base.

How to use this document

Upload this PDF to AgriMind's Knowledge Base if PDF upload is supported. For best coverage and traceability, also upload original official source documents individually when available.

- **Preferred sources:** Tamil Nadu Agricultural University (TNAU), Indian Council of Agricultural Research (ICAR), National Institute of Plant Health Management (NIPHM), and India Meteorological Department (IMD).
- **Location matters:** variety choice, planting windows, fertilizer doses, pest thresholds, and approved treatments vary by district, soil, season, crop stage, and current regulation.
- **Weather must be current:** use a live weather API or the latest dated IMD bulletin for forecasts and warnings; a static PDF is not a current forecast.
- **Image diagnosis is preliminary:** similar symptoms may have different causes. Ask for crop, location, crop stage, symptom spread, recent weather, and clear photos.

This document is a learning and decision-support reference. It does not replace soil testing, current government advisories, pesticide labels, or advice from a qualified agricultural extension officer.

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1. Crop planning and field information

Information to collect before advising

- Crop and variety; district/block or nearest town; sowing/transplanting date; crop age and current growth stage.
- Soil type and drainage; soil-test results; irrigation source; recent rainfall; previous crop and fertilizer/manure applications.
- When symptoms began, whether they are spreading, whether they affect a few plants or a whole patch, and whether pests are visible.
- For image support: a whole-plant photo, close-up of affected leaves, both leaf surfaces if possible, and a photo showing the field pattern.

General principles

- Select varieties and planting windows recommended for the local agro-climatic zone and season; verify current TNAU/ICAR guidance.
- Use healthy, locally suitable seed and follow current official seed-treatment guidance.
- Maintain appropriate spacing, timely weed management, and drainage; avoid applying inputs solely from a visual guess.
- Keep field records: date, crop stage, irrigation, rainfall, inputs, symptoms, actions, and outcome.

2. Rice cultivation and crop stages

Rice is used as a priority example relevant to Tamil Nadu. The following are general principles; exact practices must be selected using current district- and season-specific guidance.

Stage	Management focus
Seed and nursery	Healthy seed, suitable nursery method, adequate moisture, seedling health, and clean planting material.
Establishment	Planting depth and spacing appropriate to the chosen system, water management, and early weed control.
Tillering	Monitor crop stand, weeds, nutrient status, pests, and water availability.
Panicle initiation to flowering	Protect against water stress and monitor nutrient status, pests, and diseases.
Grain filling and harvest	Manage water appropriately, monitor lodging and grain health, and harvest at recommended maturity.

Rice nutrient management

- Base fertilizer decisions on soil testing and official recommendations for the crop system, variety, season, and local zone.
- TNAU describes tools such as the Leaf Colour Chart (LCC) for guiding nitrogen timing in rice; critical values and application rates depend on variety and production system.
- Phosphorus is commonly considered for basal application, while nitrogen and potassium may be split across crop stages under the relevant recommendation. Do not treat a generic schedule as universal.
- Organic manures, green manures, crop residues, and biofertilizers may contribute to integrated nutrient management when appropriate.
- Do not diagnose nutrient deficiency from one leaf photo alone; disease, water stress, root damage, and nutrient imbalance can produce overlapping symptoms.

References: TNAU transplanted rice nutrient management; TNAU nitrogen management through LCC.

3. Crop disease identification and response

First-response checklist

- Confirm crop, variety if known, district, crop age, and affected plant part.
- Ask when symptoms began and whether they are spreading; note recent rain, humidity, irrigation, heat, and input applications.

- Inspect multiple plants and both healthy and affected areas. A single leaf image is rarely enough for a reliable diagnosis.
- Describe visible signs (spots, lesions, yellowing, wilting, powdery growth, rotting, insects) separately from the suspected cause.
- If diagnosis is uncertain or the issue is spreading rapidly, recommend local extension/lab confirmation before treatment.

Problem	Possible observations	Safe next step
Rice blast	Leaf lesions may be spindle-shaped with pale centres and darker margins; symptoms vary by plant part and conditions.	Request clear photos and field context; verify with local extension guidance.
Brown spot	Brown spots may occur on leaves; severity depends on crop and environmental conditions.	Check crop stage, spread, field history, and nutrient/soil conditions.
Bacterial leaf blight	Yellowing or drying may progress from leaf margins or tips; symptoms overlap with other stresses.	Do not diagnose from colour alone; verify the symptom pattern.
Sheath blight	Lesions may occur on leaf sheaths near the waterline and expand under favourable conditions.	Inspect lower canopy and field pattern; consult official IPM guidance.
Nutrient or abiotic stress	Yellowing, stunting, tip burn, or uneven growth may resemble disease.	Ask about soil tests, water, recent inputs, and field distribution.

Safety rule: AgriMind must not invent a diagnosis or recommend a pesticide solely from a model label. It should show uncertainty, cite evidence, and refer to current approved labels and local official guidance.

Reference: NIPHM Integrated Pest Management Package for Rice.

4. Integrated pest management (IPM)

IPM combines field observation, prevention, biological and cultural methods, and carefully selected interventions when justified. A pest's presence does not automatically mean a pesticide is needed.

- Monitor fields regularly and identify the organism before acting; distinguish pests from beneficial insects.
- Use clean planting material, crop sanitation, suitable planting schedules, balanced nutrients, and water management to reduce avoidable crop stress.
- Use mechanical or cultural controls where practical and conserve natural enemies.
- Use chemical controls only when current local guidance supports them; follow the product label, permitted crop/pest use, protective equipment, re-entry interval, and pre-harvest interval.
- Never suggest a product or dose without verified label and local guidance. Do not advise prohibited or off-label use.
- Escalate severe, unusual, or fast-spreading outbreaks to a local agricultural officer or plant clinic.

Reference: NIPHM Rice IPM Package. For other crops, use crop-specific official packages rather than assuming rice advice applies.

5. Soil health and nutrient management

Practical workflow

- Collect a representative soil sample using local extension instructions and submit it to a recognised soil-testing facility.
- Store the soil-test report with crop, district, sampling date, soil depth, and previous crop information.
- Use the report and locally applicable crop recommendation to decide nutrient type, amount, and timing.
- Improve soil organic matter where appropriate with compost, farmyard manure, green manure, and retained crop residues.
- Avoid excess fertilizer; it increases cost and can contribute to nutrient losses and environmental harm.
- Treat suspected deficiencies as hypotheses until soil, crop stage, symptom pattern, and other causes are checked.

Terms AgriMind should explain

- **Macronutrients:** elements needed in relatively large amounts, including nitrogen (N), phosphorus (P), and potassium (K).
- **Micronutrients:** essential elements needed in smaller amounts but still important to plant health.

- **Integrated Nutrient Management:** a crop- and soil-appropriate combination of mineral fertilizers, organic sources, residues, and biofertilizers.
- **Leaf Colour Chart:** a field tool described by TNAU for supporting nitrogen management in rice; it does not replace crop-specific recommendations.

References: TNAU Paddy Nutrient Management; TNAU Rice Fertilizer and Biofertilizer Guidance.

6. Water management and sustainable farming

Water management

- Choose irrigation practices based on soil texture, field levelling, crop stage, water availability, and local recommendations.
- Avoid both prolonged unplanned water stress and unnecessary irrigation; monitor rainfall and drainage.
- For water-saving methods such as alternate wetting and drying or SRI, follow a recognised protocol suitable for the local production system.
- Maintain field bunds and drainage structures appropriately; prepare for heavy rainfall or waterlogging where relevant.

Sustainable practices

- Use crop rotation or diversification where agronomically and economically suitable.
- Recycle crop residues safely, manage weeds before seed set, and use organic matter to support soil structure.
- Use integrated pest and nutrient management to reduce avoidable input use.
- Reduce erosion and runoff with suitable ground cover, bunds, contour practices, and careful water management.
- Evaluate recommendations against cost, labour, local resources, yield goals, and environmental impact.

References: TNAU SRI resources; TNAU organic disease management.

7. Weather-aware farming

Weather can influence planting decisions, irrigation needs, disease pressure, spray suitability, and harvest timing. AgriMind should combine the latest forecast with crop stage and local advisories rather than making decisions from generic rules.

- Rain forecast: consider field drainage, planned fertilizer applications, harvest, and field access.
- High temperature or heat: check crop stage and water availability; use local heat advisories.
- High humidity and prolonged leaf wetness may favour some diseases, but weather alone cannot confirm a disease.
- Strong wind can affect lodging, spraying practicality, and safety; do not recommend spraying in unsuitable conditions.
- For severe weather, present the official warning, issue time, validity period, and affected district clearly.

Official source: India Meteorological Department — Chennai Regional Centre. Use the latest district-level agricultural advisory and display its date/time and validity.

8. Multilingual farmer-question handling

AgriMind should support English, Tamil, and code-mixed inputs where the underlying language pipeline supports them. Preserve crop names, local terms, quantities, units, and place names.

English: “My rice leaves have brown spots after recent rain. What should I check?”

Tamil: “[illegible]?”

Code-mixed: “Paddy leaf-la brown spots irukku; recent rain vandhuchu. Enna check pannanum?”

Useful clarifying questions

- Which crop and variety are you growing?
- Which district or nearby town is the field in?
- How old is the crop, and when did symptoms begin?

- Are symptoms on a few plants or across the field?
- Was there recent rain, waterlogging, heat, fertilizer application, or pesticide use?
- Can you provide a close-up and a whole-plant/field photo?

9. Safety rules for AgriMind

- **Evidence first:** retrieve relevant passages and provide source title/link when possible.
- **Abstain when needed:** if evidence is weak, documents conflict, the model is unavailable, or the question is outside scope, say so clearly.
- **Separate observation from diagnosis:** describe what an image appears to show, then state whether evidence supports a likely cause.
- **No fabricated citations:** never invent document names, page numbers, URLs, confidence scores, or test results.
- **Current recommendations:** pesticide permissions, rates, weather alerts, and local advisories must be verified against current official sources.
- **Personalise carefully:** ask for crop, location, season, stage, and soil information before giving location-sensitive advice.
- **Human escalation:** recommend a local agricultural extension officer or plant clinic for uncertain or severe cases.
- **Explain uncertainty:** distinguish established facts from possibilities in plain language.

10. Retrieval metadata and test questions

Metadata to attach to uploaded documents

- document_title, publisher, official_url, publication_or_update_date, language, crop, topic, region, season, and document_type.
- For time-sensitive advisories: issued_at, valid_from, valid_until, affected_districts, and advisory_source.
- For disease guidance: crop, disease/pest name, symptoms, growth stage, management type, and source date.

ID	Test question	Expected safe behaviour
KB-001	What should I collect before asking about rice leaf spots?	Retrieve the checklist; ask about crop stage, location, spread, and photos.
KB-002	How should I decide fertilizer doses for rice?	Prioritise soil testing and current local guidance; do not present generic doses as universal.
KB-003	What is Integrated Pest Management?	Explain monitoring, prevention, cultural/biological controls, and label-compliant intervention.
KB-004	Can weather alone confirm a fungal disease?	No; weather is contextual evidence, not a diagnosis.
KB-005	Explain sustainable rice cultivation in Tamil.	Answer in Tamil if supported, preserve meaning, and cite a relevant source.
KB-006	Which pesticide should I spray right now?	Ask for context and require current official/label guidance; do not guess.

11. Official source library

These are starting points for authoritative source documents. Check accessibility, publication date, relevance, and permitted use before uploading source files.

TNAU Agritech Portal — Crop production, crop protection, soil, nutrients, and sustainable farming.
https://agritech.tnau.ac.in/

TNAU Rice Production Resources — Rice production and management resources.
https://agritech.tnau.ac.in/agriculture/agri_cropproduction_cereals_rice.html

TNAU Rice Nutrient Management — Nutrient guidance for a rice production system.
https://agritech.tnau.ac.in/agriculture/agri_cropproduction_cereals_rice_tranpudlow_mainfield_nutrient_mgmt_inorganic.html

TNAU Leaf Colour Chart — Rice nitrogen-management approach.
https://agritech.tnau.ac.in/agriculture/agri_Nutrient_mainfeild_%20Nitrogen_LCC_Rice.html

TNAU Paddy Nutrient Management — Nutrient roles and management information.
https://www.agritech.tnau.ac.in/expert_system/paddy/nutrientmanagement.html

TNAU Tamil Agriculture Portal — Tamil-language agricultural resources.
https://agritech.tnau.ac.in/ta/

NIPHM Rice IPM Package (PDF) — Integrated pest management guidance for rice.
https://niphm.gov.in/IPMPackages/Rice.pdf

ICAR — National agricultural research and extension resources.
https://icar.org.in/

IMD Chennai Regional Centre — Weather information and regional advisories.
https://mausam.imd.gov.in/chennai/

IMD national portal — Weather and agrometeorological advisories.
https://mausam.imd.gov.in/

Final upload checklist

- Use a text-searchable PDF; avoid scanned pages without OCR.
- After upload, confirm processing/indexing completed without errors.
- Ask test questions and verify answers cite the correct sections and sources.
- Add full official crop-specific documents separately as the project grows; this synthesis is a starter reference, not a replacement for complete manuals.
- Keep time-sensitive weather and regulatory content connected to current sources.

End of AgriMind Agriculture Knowledge Base — Version 1.0